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Classification

Definition: Classification is the process to group or arrange data separately in aspect of their common characteristics. After collection and editing of data an important step towards processing the data is classification.

Example: It is well known that the letters collected in a post office are sorted into different lots or a geographical basis, i.e., in accordance with their destinations as Dhaka, Chittagong, Rajshahi, Khulan etc.

Characteristics of classification

- a) Classification performs homogeneous grouping of data
- b) It brings out points of similarity and dissimilarities.
- c) The classification may be either real or imaginary
- d) Classification is flexible to accommodate adjustments

Objectives / purposes of classifications

- a) To simplify and condense the large data
- b) To present the facts to easily in understandable form
- c) To allow comparisons
- d) To help to draw valid inferences
- e) To relate the variables among the data
- f) To help further analysis
- d) To eliminate unwanted data
- g) To prepare tabulation

Guiding principles (rules) of classifications

Following are the general guiding principles for good classifications

- a) **Exhaustive:** Classification should be exhaustive. Each and every item in data must belong to one of class. Introduction of residual class (i.e. either, miscellaneous etc.) should be avoided.
- b) **Mutually exclusive:** Each item should be placed at only one class
- c) **Suitability:** The classification should confirm to object of inquiry.
- d) **Stability:** Only one principle must be maintained throughout the classification and analysis.
- e) **Homogeneity:** The items included in each class must be homogeneous.
- f) **Flexibility:** A good classification should be flexible enough to accommodate new situation or changed situations.

Types of Classification

Data can be classified on the following four bases:

- (1) **Geographical Classification** → i.e., area-wise, e.g., cities, districts, roads etc. Geographical classifications are usually listed in alphabetical order for easy reference.
- (2) **Chronological Classification** → i.e., on the basis of time. When data are observed over a period of time (such as years, months, weeks, days, hours etc.) the type of classification is known as chronological classification. Time series are usually listed in chronological order normally starting with the earliest period. For example National income is expressed every year and departmental sales are expressed every month or week.
- (3) **Qualitative Classification** → i.e., according to some attributes. In qualitative classification, data are classified on the basis of some attributes or qualities such as sex, color of eye, color of hair, literacy, nationality, religion, etc.
- (4) **Quantitative Classification** → i.e., in terms of magnitudes. Quantitative classification refers to the classification of data according to some characteristics that can be measured, such as height, weight, income, sales etc.

Tabulation

Definition: Tabulation is the process to arrange the data according to their characteristics in separate columns vertically and rows horizontally. The purpose of a table is to simplify the presentation, and to facilitate comparisons. For example

TABLE_PRODUCT

Product ID	Color	Price
1	red, green	15.99
2	yellow	23.99
3	green	17.50
4	yellow, blue	9.99
5	red	29.99

Major Objectives of Tabulation

1. To simplify the complex data
2. To facilitate comparison
3. To economies the space
4. To draw valid inference / conclusions
5. To help for further analysis

Steps/Parts/Rules of Tabulation

The following steps/parts/rules are needed for constructing tabulation:

- (1) **Table Number:** For more than one, tables should be numbered.
- (2) **Date:** Table may be dated at time of completion.
- (3) **Title:** Each table must have a suitable and short title describing the contents of the table.
- (4) **Master Caption:** Master caption refers to the vertical data of all columns through homogeneous variables.
- (5) **Column Heading:** Column heading is the basic title of similar ranges of data set.
- (6) **Sub-column Heading:** It divides the column heading in different groups.
- (7) **Column Number:** Each column may be numbered for explanations.
- (8) **Unit Head:** It refers to the measurement units for each column or row.
- (9) **Cell Insides:** It is optional. It refers each cell may include an inside for alternative frequencies.
- (10) **Stub:** Stubs are the designations of row headings.

- (11) **Sub-stub:** Stubs may be divided into different sub-rows called sub-stubs.
- (12) **Body:** Body is the entire statistical data set.
- (13) **Head Notes:** It is the explanatory statement of the data set placed below the title of the table and may enclosed by brackets.
- (14) **Foot Notes:** Difficult items in the table are explained for better understanding which placed below the table.
- (15) **Source Notes:** Data sources are listed below the footnotes.
- (16) **References:** References are listed below the source notes.

Differences between Classification and Tabulation

1. First data are classified and presented in tables; classification is the basis for tabulation.
2. Tabulation is a mechanical function of classification because is tabulation classified data are placed in row and columns.
3. Classification is a process of statistical analysis while tabulation is a process of presenting data is suitable structure.